Database

Without database :

→ Storing data in text files, spreadsheets, etc.

→ which is a very slow process and retrieving data from a large, unstructured collections is error-prone, and inefficient.

Eg: Searching for a specific customer's order in 10,000 Excel files (takes hours)

With database

→ Organize the collection of data in the form of table, graph, key-value pair, etc,..

→ The task of storing, managing and retriving the data becomes a easy task using tools like mySQL

Eg: Find the same customer's order in DB take less than a min with simple query

There are two different type of database

→ Relational database (structured i.e table)

→ Non-Relational databe (unstructured i.e graph, doc)

Relational database: Stores data in the form of row and column, and have relationship among various table

Eg: MySQL, SQL Server, PostgreSQL, etc..

Interview Question:

There are different types of databases right? There are Oracle, MySQL, MongoDB, SQL server, postgreSQL.

What is the difference b/w these databases?

Non-Relational database:

Store data in the form of grap, key-value pair, doc.

Eg: HBase, MongoDB, Cassandra, etc

There are two different type of database

→ Relational database (structured i.e table)

→ Non-Relational datebe (unstructured i.e graph, doc)

Relational database: Stores data in the form of row and column, and have relationship among various table

Eg: MySQL, SQL Server, PostgreSQL, etc.

Interview Question:

There are different types of databases right? There are Oracle, MySQL, ~~[illegible]~~ MongoDB, SQL server, postgreSQL.
What is the difference b/w these databases?

Non-Relational database:

Store ~~[illegible]~~ data in the form of grap, key-value pair, doc.

Eg: HBase, MongoDB, Cassandra, etc

I/Q Ans:

→ There are different types of database like oracle, MySQL, Mongo DB, etc. The main difference is that relational database, non-relational database.

→ where MySQL, Oracle comes under RDB then MongoDB comes under NRDB.

Data → refers collection of raw facts or fig
Eg: XYZ, organge

Information → Processed data is called information → orgnge colour, orange fruit

↳ it gives meaning to a unprocessed data

SQL

MYSQL database → to handle the MySQL database, we need Query language called SQL.

→ MYSQL is a tool on which we are writing a SQL query to fetch the data, store the data and also to manage the data.

Practice: GoormIDE platform [cloud based platform]
MYSQL workbench

COMMANDS

CD1 SHOW DATABASES; (show all the available databases in the DBMS software)

CD2 CREATE DATABASE {techForAllWithMonish};
(this command is used to create a new database)

CD3 CREATE DATABASE IF NOT EXISTS {techForAllWithMonish}

(it only create database ~~if~~ if it is not available in MySQL platform)

→ This query is used because, if we try to create the already existing DB with the command CREATE DATABASE {techForAllWithMonish}

Error → It will give error as DB already exists

→ if we use IF NOT EXISTS command it will give warning

CD4 USE techForAllWithMonish;
(to choose particular database)

CD5 SELECT DATASE();
(to identify the current DB that we are working)

[After choosing the DB only we can use CD5]

CRUD operation

- create: a new table
- Read: the data inside the table
- Update: the data
- Delete: instance, table, dB

Creating a new table

INT
VARCHAR

SYNTAX:

```
CREATE TABLE {table_name} (
 col 1  datatype,
 col 2  datatype,
  :
);
```

Eg

```
CREATE TABLE employee (
 FirstName varchar(30),
 LastName varchar(30)
 Location varchar(20)
 Salary int
);
```

to get Description of table

SYNTAX:

DESC {table_name}

Eg:

Desc employee;

O/P

Field	Type	Null	Key	Default	Extra
FN	varchar(30)	YES		NULL	
LN	(40)	YES		NULL	
L	(20)	YES		NULL	
S	int	YES		NULL	